

ETHEREUM ECOSYSTEM GRANT APPLICATION

Ethereum-native civic infrastructure

21 lets anyone report a real-world problem, fund the fix, hire the people who do the work, and verify the result — settled on Ethereum.

ASK	STAGE	LICENSE
$75,000 / 10 mo.	MVP + 3 city pilots	Open source · MIT

01 — THE PROBLEM

Cities can see the problem. They cannot coordinate the fix.

01 **Reporting ends in a ticket.**

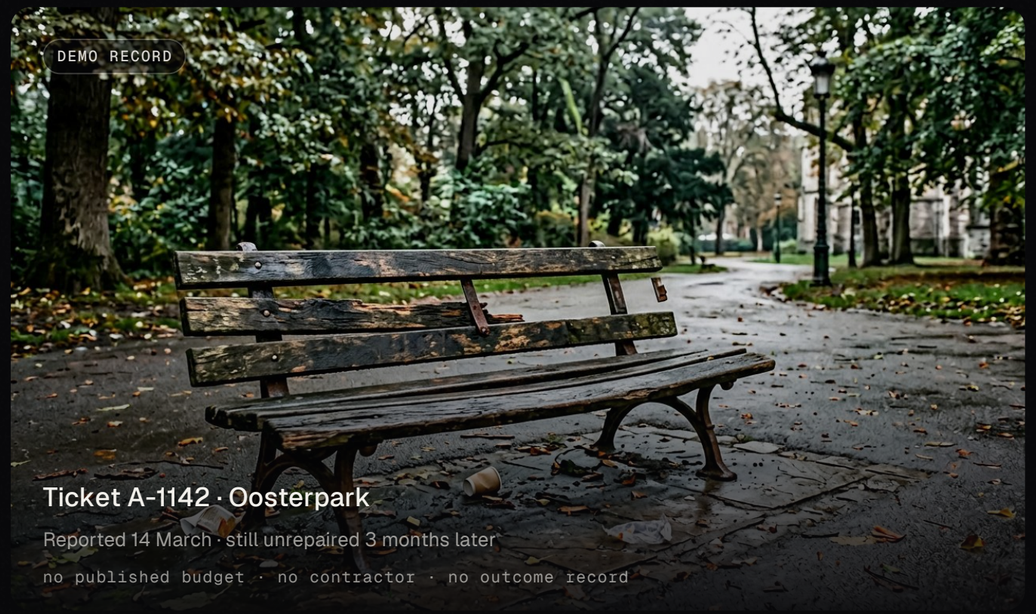
A photo to a hotline produces a reference number, not a budget. Residents never learn who decided, who was paid, or what actually happened.

02 **Small works fall through.**

A €420 bench, a €950 bike rack, a €300 mural: too small for procurement, too visible to ignore. They wait for a season that never comes.

03 **Public-goods funding stops at the screen.**

Crypto has capital, matching pools and governance tooling for digital work. There is no delivery layer for physical work.



WHY THIS IS BUILDABLE NOW

- **L2 fees** — a €4 settlement is economically sane
- **Personhood** — proof-of-personhood is usable
- **Attestations** — EAS makes verified claims portable
- **Funding** — QF + retro funding is mature

A public protocol for getting real things done.

21 turns a geotagged report into a funded, contracted and verified piece of work. The money moves through an on-chain escrow; the decision, the evidence and the payout stay public.

01 **Report**
Photo, location, category, urgency

02 **Propose**
Contractors publish budget + milestones

03 **Fund**
Community + matching pool fill escrow

04 **Decide**
Eligible participants select a proposal

05 **Build**
Work happens in the physical world

06 **Verify**
Evidence + quorum of attestations

07 **Settle**
Tranche released to the contractor

Non-custodial
21 never holds funds. A per-task escrow contract does.

Rule-bound
Verification and payment rules are pinned before funding opens.

Composable
Any city, DAO or funder can plug into the same registry.

Forkable
MIT-licensed contracts and client. No lock-in for municipalities.

SAME REGISTRY, MANY FRONT DOORS

A city can run its own branded instance, a funder can open a round, a contractor can use a mobile-only client — all writing to one task registry.

03 — ONE TASK, END TO END

A broken bench in nine days.

Demo scenario built from the product spec. Amounts and dates are illustrative, not reported results.



Before — three missing slats, reported from the phone



After — replaced and re-fixed by Groenwerk BV

TASK LEDGER · PUBLIC BY DEFAULT

Day 0	Task published	€420 target
Day 1	3 proposals received	€380 - €455
Day 3	84 residents voted	62% Groenwerk
Day 4	Escrow funded	€420 in USDC
Day 7	Milestone evidence	6 photos + GPS
Day 9	Dual attestation	payment released

WHAT THE RESIDENT GETS

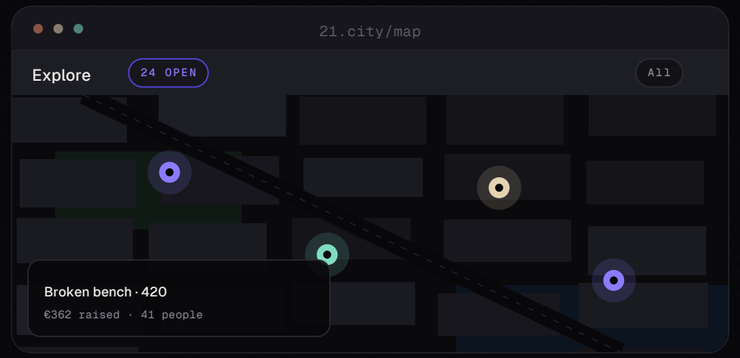
A public URL with the full history, the contractor's identity and the final payment.

WHAT THE CONTRACTOR GETS

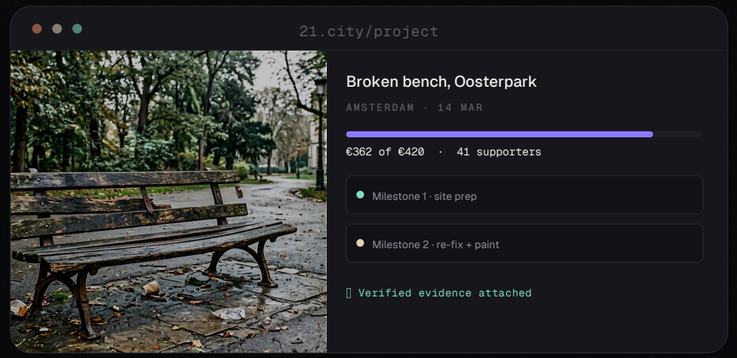
Milestone payments released in stablecoin the day verification clears — no 90-day invoice cycle.

Six surfaces, one continuous workflow.

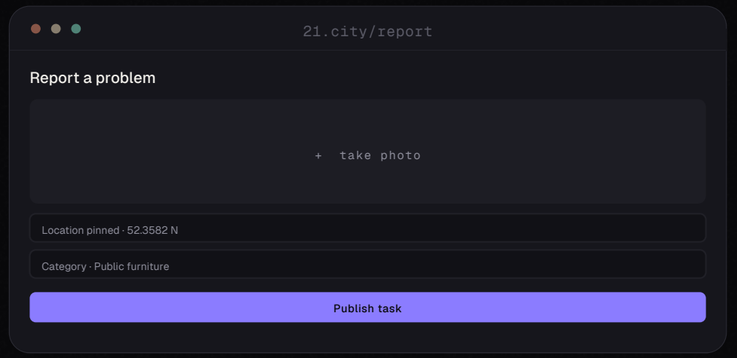
Responsive web app plus an installable mobile PWA. Every screen below is part of the MVP scope, not a future release.



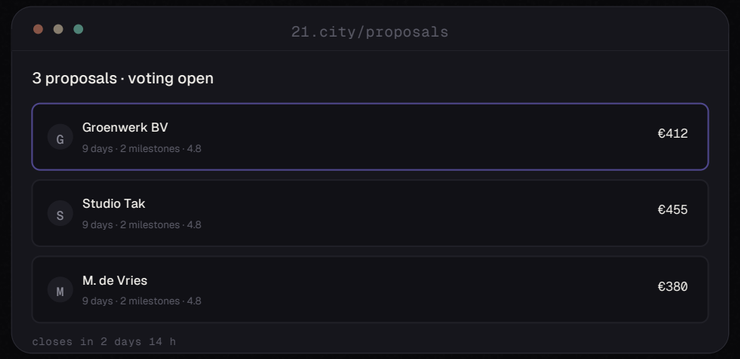
01 **Explore** Map-first discovery with live task markers



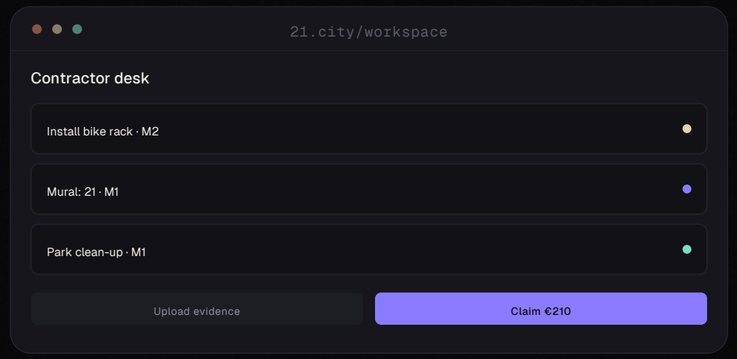
02 **Project** Evidence, budget, milestones, discussion



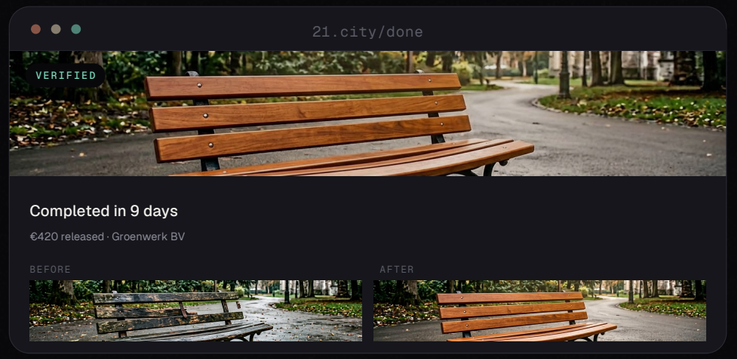
03 **Report** Photo, GPS, category — under 60 seconds



04 **Proposals** Competing bids with itemised budgets



05 **Contractor** Bids, contracts, evidence, payouts



06 **Completed** Before / after and the payment record

Money moves only when the street agrees.

A photo alone is not proof. 21 combines device-signed capture, geofencing, independent attestors and an economic challenge window.

01 Device-signed capture

Photos are taken in-app: EXIF, GPS, gyroscope and a hardware-backed signature are hashed and stored with the file.

02 Proximity, not claims

The capture must fall inside the task geofence; high-value tasks require a reviewer visit.

03 Dual attestation

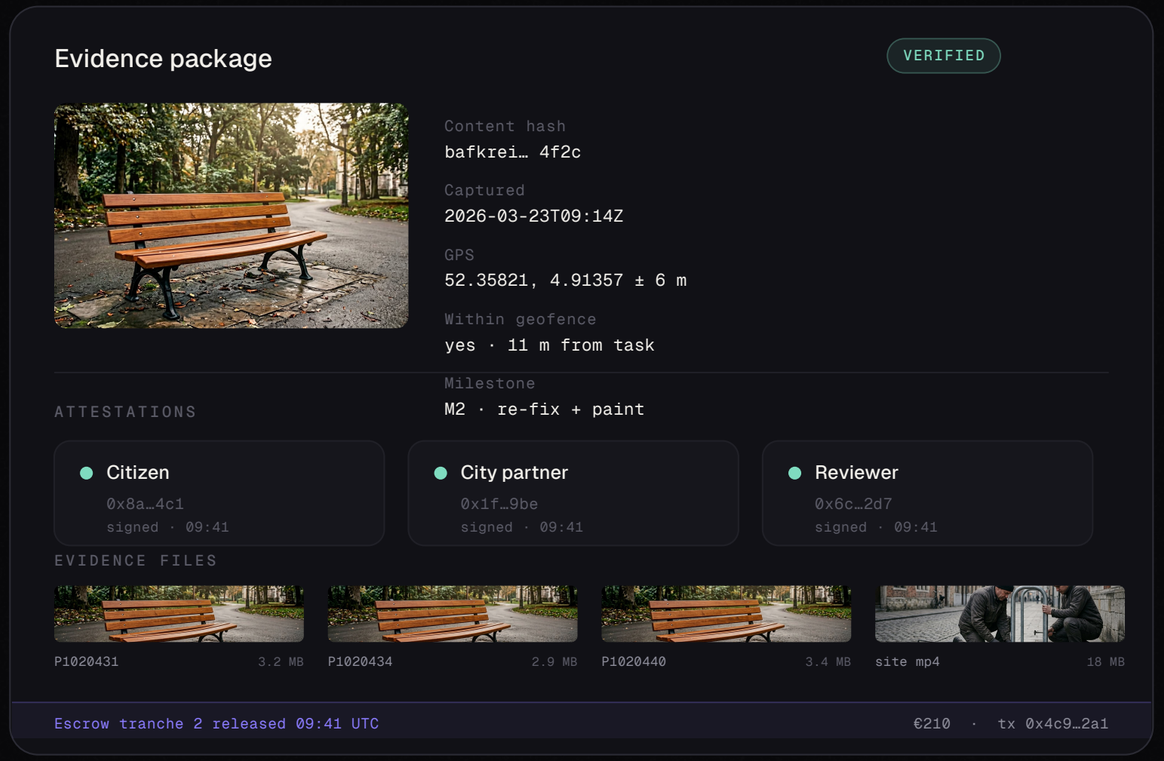
A citizen attestor and a city partner sign the milestone. A random reviewer is added above €1,000.

04 Challenge window

72 hours to dispute with a staked bond. Losing a dispute costs the bond; winning returns it plus a share of the fee.

05 Reputation that cannot be bought

Scores come from verified work only (ERC-5192, non-transferable) and decay without activity.



Funding buys outcomes, not votes.

Each project publishes its rules before the first euro arrives. The rules hash is pinned on-chain, so the game cannot change mid-round.

01

Who may vote

Three configurable profiles

- Resident: proof-of-personhood inside a 2 km radius
- Contributor: any wallet above a €5 floor
- Partner: city wallet, advisory weight only

02

How money is weighted

Default profile: one person, one vote

- A €5 and a €500 contribution carry equal voice
- Token-weighted modes require an explicit opt-in by the city
- Quadratic matching from the 21 pool, capped per project

03

How a winner is chosen

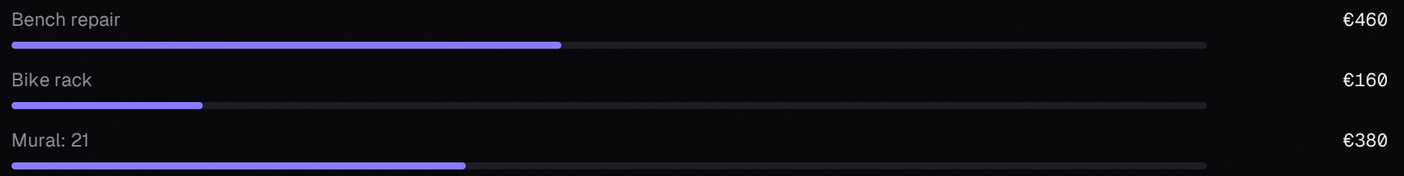
Approval vote, 5-day window

- Quorum: 25% of eligible voters, minimum 20 people
- No quorum → partner + council decide, rationale published
- Losers' proposals stay open for other rounds

QUADRATIC MATCHING, ILLUSTRATED

Pool €1,000 · 3 projects

Every contribution is square-rooted before it counts, so twenty small donors outrank one large one.



RULES HASH `0x9d2e71…c40b` · pinned at funding open · quorum 25% · 5-day vote · one-person-one-vote — editable only with 72h timelock

21 is a number anyone can paint.

A commissioned, always-authorised mural programme runs alongside the protocol. It is how a civic tool becomes recognisable on a street in Lagos or Lisbon without a translation.

01 **Commission, never tag**

Every wall has a signed owner consent, a removal clause and a paid artist. The registry refuses locations without a permission record.

02 **21 murals in 21 cities**

Year one: €300–€1,200 per wall, funded by the campaign pool and local sponsors. Each is a Proof-of-Place entry on-chain.

03 **The number is the message**

No slogan to translate, no logo to explain. Two digits, painted by hand, pointing back to a working protocol.



21	€300–1,200	100%	1
cities in year one	artist fee per wall	consented walls	permanent registry entry per mural

Serious enough for a treasury. Simple enough for a phone.

A compact system ships with the MVP: tokens, components, interaction states and usage rules, documented in a public Storybook.

WORDMARK

21

Build the world together.

Lockup: app icon · sticker · wall · favicon

Geist Sans + Geist Mono · OFL · no gradients, no glow

PALETTE

- Paper #F3F1EC
- Graphite #16161C
- Signal violet #8B7CFF
- Deep violet #5343D6
- Verified #7FDCC0
- Sand #E4D3B2

TYPE SCALE

Display 56 / 60

Title 30 / 36

Body 13 / 19

LABEL 8.5 / MONO / 1.5

COMPONENTS

REPORT AN ISSUE | EXPLORE MAP | VERIFIED | VOTING OPEN | PENDING · 12s | INSUFFICIENT FUNDS

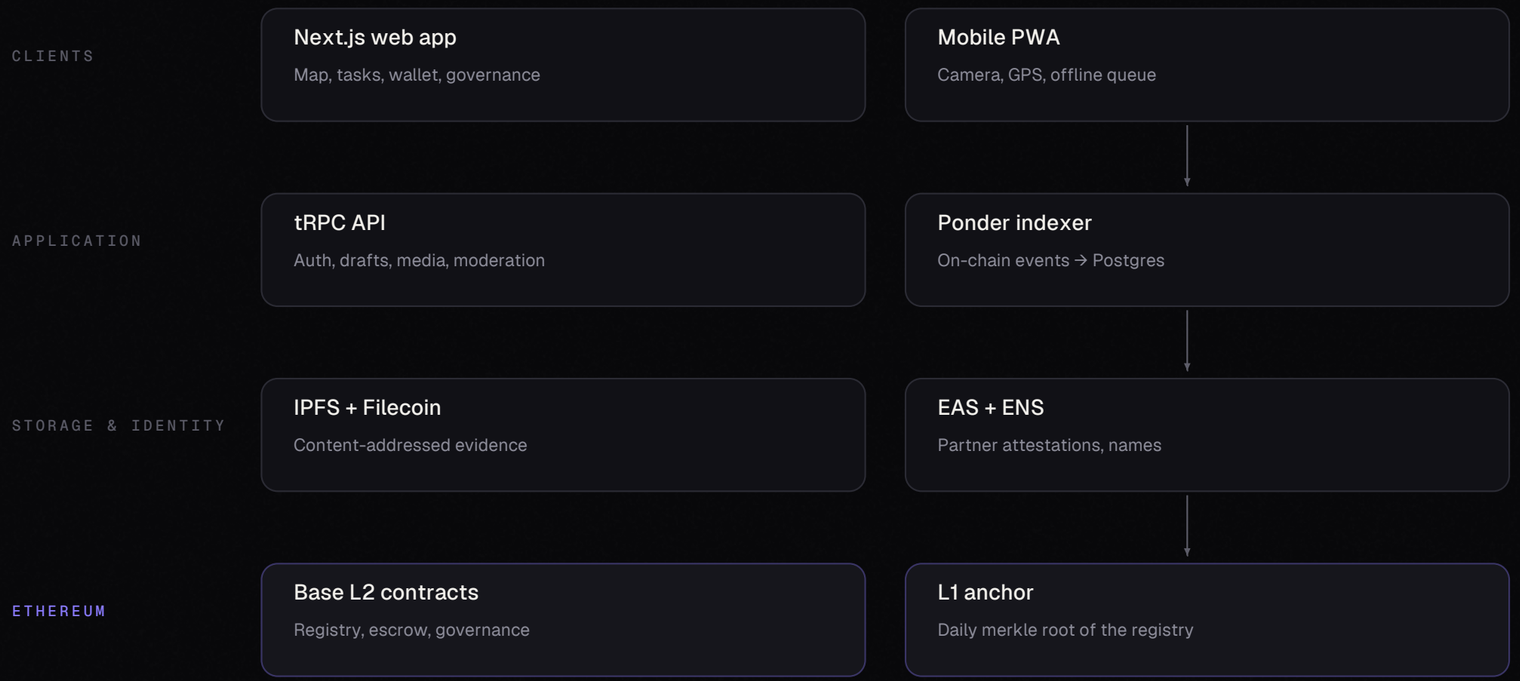
LIFECYCLE

draft → proposals → voting → funding → in progress → verification → settled

one colour, one word, one next action per state

Boring where it matters. Verifiable where it counts.

TypeScript end to end, Foundry for contracts, no custodial backend.



CONSTRAINTS

Gas per task
< $0.02 on Base

Contract standard
Solidity 0.8 / Foundry

Upgrade policy
Immutable core + factory

Data on-chain
State + hashes, never media

Who can pause
3-of-5 guardian, 48h delay

TRUST BOUNDARY

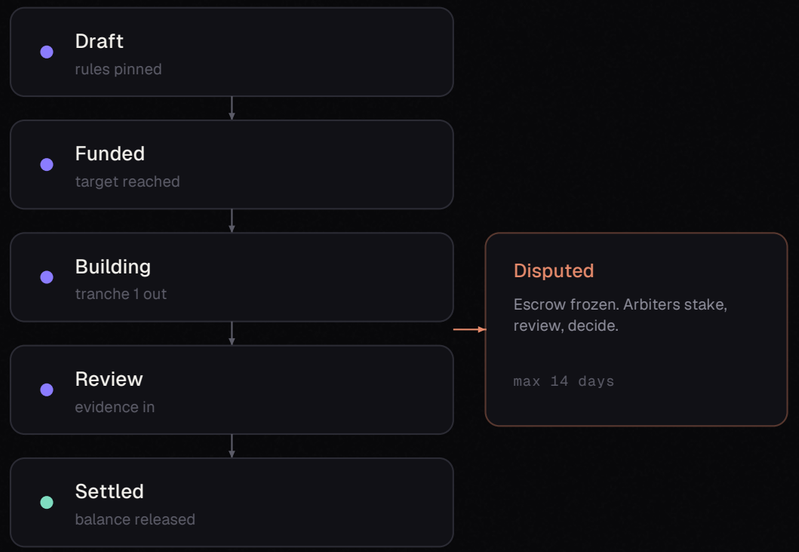
Funds live in escrow contracts. The 21 server can censor a list, but it cannot move, freeze or redirect a single euro.

Eight small contracts, one escrow per task.

Immutable core deployed by a factory, so each task gets its own escrow address and its own accounting. No proxy admin can touch user funds.

`TaskRegistry`	Canonical task record: geohash, category, media CID, status
`ProposalBook`	Bids, itemised budgets, milestone schedule, withdrawal
`FundingPool`	Contributions, refunds on failure, matching accounting
`Escrow`	Per-task USDC, milestone tranches, release conditions
`Verification`	Attestation quorum, evidence CIDs, challenge window
`Dispute`	Staked arbiters, slashing, escalation to council
`Reputation`	Non-transferable score from verified completions
`Treasury`	Matching pool, Safe multisig + 72h timelock

ESCROW LIFECYCLE



Escrow rules are hashed and emitted on TaskPublished. A task cannot be funded until its payment logic is immutable.

The attacker is a person with a camera.

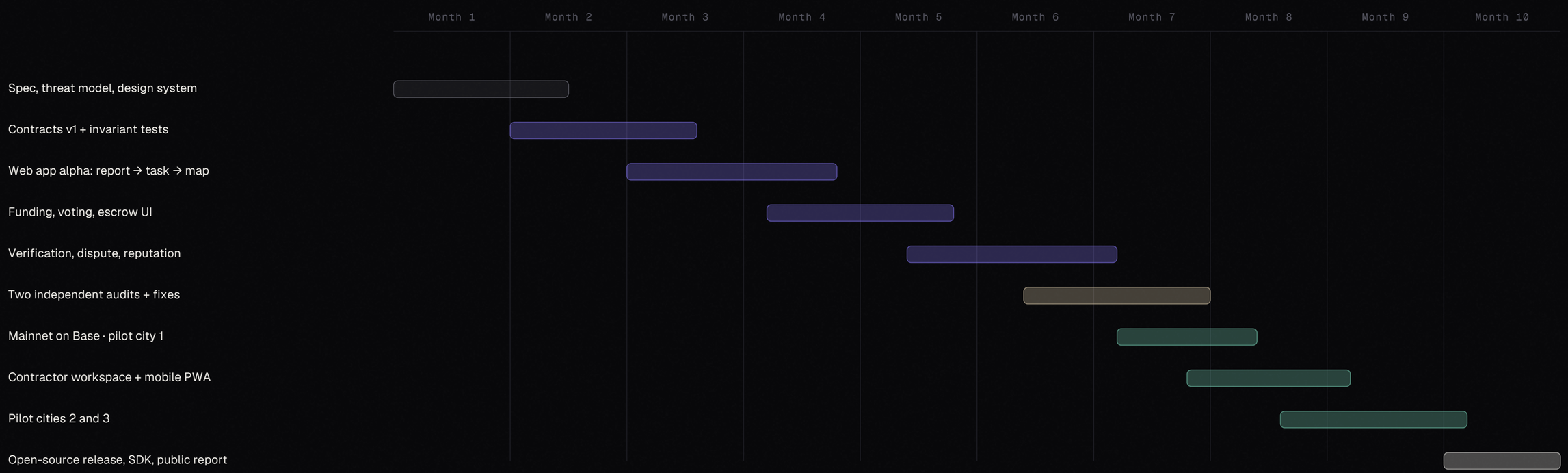
Physical-world verification has adversarial failure modes that a normal audit does not catch. Each one is handled explicitly.

THREAT	WHY IT IS REAL	MITIGATION IN THE MVP
Fabricated or AI-generated proof	A model can draw a fixed bench in seconds.	In-app capture only, signed device metadata, C2PA content credentials, phash dedupe, human spot audits.
GPS spoofing	Mock locations are one developer setting away.	Geofence + cell/Wi-Fi corroboration, on-site reviewer above €1,000, anomalous-radius flags.
Reporter–contractor collusion	The easiest way to extract money from a pool.	Role separation, random reviewer panel, reputation at risk, payout caps for new addresses.
Sybil voting	Free wallets, cheap votes.	Personhood-gated eligibility, snapshot at vote open, one-person-one-vote as the default.
Escrow bug or exploit	Contracts hold real money on day one.	Two independent audits, invariant fuzzing, per-task caps, 3-of-5 guardian pause with 48h delay.
Privacy of residents	Photos contain faces, plates and windows.	On-device blur, coarse public coordinates, off-chain personal data, GDPR DPIA before pilot 1.

Open question we bring to reviewers: what is the minimum viable proof for a €300 repair that still resists a determined faker?

Ten months from spec to three live cities.

Every milestone ships something a reviewer can open, test or audit. Nothing is marked done without a public artefact.



RELEASE GATES

Month 6 audits closed · no critical or high findings open
milestone M5

Month 7 first real euro spent through escrow
milestone M6

Month 10 three pilots, public report, open SDK
milestone M8

What the grant actually pays for.

Acceptance criteria are written as artefacts, not intentions. Each milestone is reviewed before the next tranche is released.

	MILESTONE	ACCEPTANCE CRITERION	SHARE
M1	Protocol spec + threat model	Published RFC, 2 external reviews	5%
M2	Contracts on testnet	90%+ coverage, invariants fuzzed, testnet deploy	10%
M3	Alpha app: report → task → escrow	A bench task completed end-to-end on testnet	15%
M4	Verification + dispute modules	Dual attestation and a resolved dispute in a live drill	10%
M5	Two independent audits	Reports published; all critical/high fixed	20%
M6	Base mainnet launch	Verified contracts, per-task cap, guardian live	10%
M7	Three city pilots	3 partners, 20 tasks each, real payouts	20%
M8	Open source + SDK	MIT release, docs, 3 external integrators	10%

Cumulative release: 15% by M2, 40% by M4, 70% by M6, 100% at close-out. An unmet criterion means an unpaid milestone, not a renegotiation.

14 — BUDGET

$75,000, allocated line by line.

People, audits and pilots. No token, no marketing spend, no margin hidden in the total.



Total $75,000 · 10 months · 2 FTE + 1 part-time

LINE ITEM	SHARE	USD
Smart contracts, tests, two audits	32%	$24,000
Product, frontend, mobile PWA	24%	$18,000
Backend, indexer, storage pipeline	13%	$9,500
Verification, dispute and reviewer tooling	10%	$7,500
Three city pilots and the mural programme	12%	$9,000
Design system, documentation, Storybook	5%	$4,000
Legal, GDPR review, operations, travel	4%	$3,000

If the grant is smaller than requested, the audits and the mainnet launch are the last things we cut.

15 — TEAM

Three roles, named before the first tranche.

21 is a small team plus one municipal advisor. Replace the bracketed fields with names and profile links before submission.

P

Protocol lead

contracts

[name] · [github / linkedin]

Solidity and Foundry, escrow design, one prior public-goods deployment. Owns M1–M6 and both audits.

commitment: full time, 10 months

P

Product & design lead

product

[name] · [github / linkedin]

Civic interfaces, map and form UX, accessible design systems. Owns M3, M7 and the pilot onboarding.

commitment: full time, 10 months

F

Full-stack engineer

backend

[name] · [github / linkedin]

Next.js, indexer, media pipeline, PWA capture and offline queue. Owns M3 and M7.

commitment: full time, 10 months

ADVISOR

Municipal procurement + public-art permissions

A former city procurement officer reviews pilot contracts, consent templates and the mural programme before any wall is touched. [name · organisation]

HOW WE WORK

- Public roadmap, public repo, weekly demo
- Every payment in milestone tranches
- No token, no presale, no paid listings

OPEN ROLE FUNDED BY TRANCHE 2

Verification & field-operations lead — runs reviewer panels in the three pilot cities.

NOT APPLYING FOR

Marketing budget, token liquidity, or exchange listings.

16 — SUCCESS METRICS

Targets, not traction.

21 is pre-launch. These are the numbers we commit to reporting at close-out, with the methodology published alongside them.

3
city partners under a signed pilot agreement
source: signed pilot MOUs

60
tasks published, verified and settled
source: registry events

€45k
value moved through escrow
source: escrow contract balance

25
contractors paid, median payout < 24h
source: payout timestamps

21
commissioned murals registered on-chain
source: proof-of-place entries

500
unique participants with proof-of-personhood
source: eligible voter snapshot

GUARDRAIL METRICS — WE FAIL IF THESE MOVE THE WRONG WAY

< 1% of released value	**< 8% of tasks**	**< 72h median**	**> 40%**
Fraud rate	Dispute rate	Time to first proposal	Repeat contractors

Because the alternative is a database someone else controls.

A civic platform that holds public money should not be able to rewrite its own history. Ethereum gives 21 four properties no SaaS can offer.

01 Neutral settlement

No company sits between a resident's euro and a contractor's invoice. The escrow is the counterparty.

02 A global payment rail

USDC on Base pays a contractor in Nairobi or Nijmegen in seconds, without a bank account.

03 Composable public goods

Quadratic funding, retro funding, EAS attestations and Safe treasuries plug in unchanged.

04 Permanent public record

What was promised, funded and delivered stays queryable — even if 21 disappears.

WHAT 21 DELIBERATELY DOES NOT DO

- × No token sale and no speculative asset
- × No custody of user funds at any point
- × No paid placement of projects or contractors
- × No growth metrics that are not on-chain verifiable

WHERE THE GRANT FITS

- 21 is public-goods infrastructure: an open registry and escrow any city, DAO or funding round can build on.
- Public-goods rounds can fund tasks directly through the same escrow — no new rails, no new audit surface.
- The grant funds the protocol and the first three proofs, not a company's growth.

Built to outlive the funding round.

The protocol is open source and unowned. Revenue, if any, is optional and capped; the public instance stays free for residents.

01

Sustainable by design

- Optional 1.5% fee on released escrow, capped and opt-in per city
- Hosted instance + support subscription for municipalities
- Matching pools funded by partners, not by the protocol

02

Progressive decentralisation

- M1–M6: core team ships, guardian multisig holds pause rights
- M7–M10: council elected by contributors, parameters on-chain
- Year 2: registry ownership renounced; fee switch under token-less vote

03

What stays open

- Contracts, indexer and client under MIT
- Public data export: every task, attestation and payment
- A documented schema so other cities can fork without us

IF THE PILOT WORKS, THE NEXT EIGHTEEN MONTHS LOOK LIKE

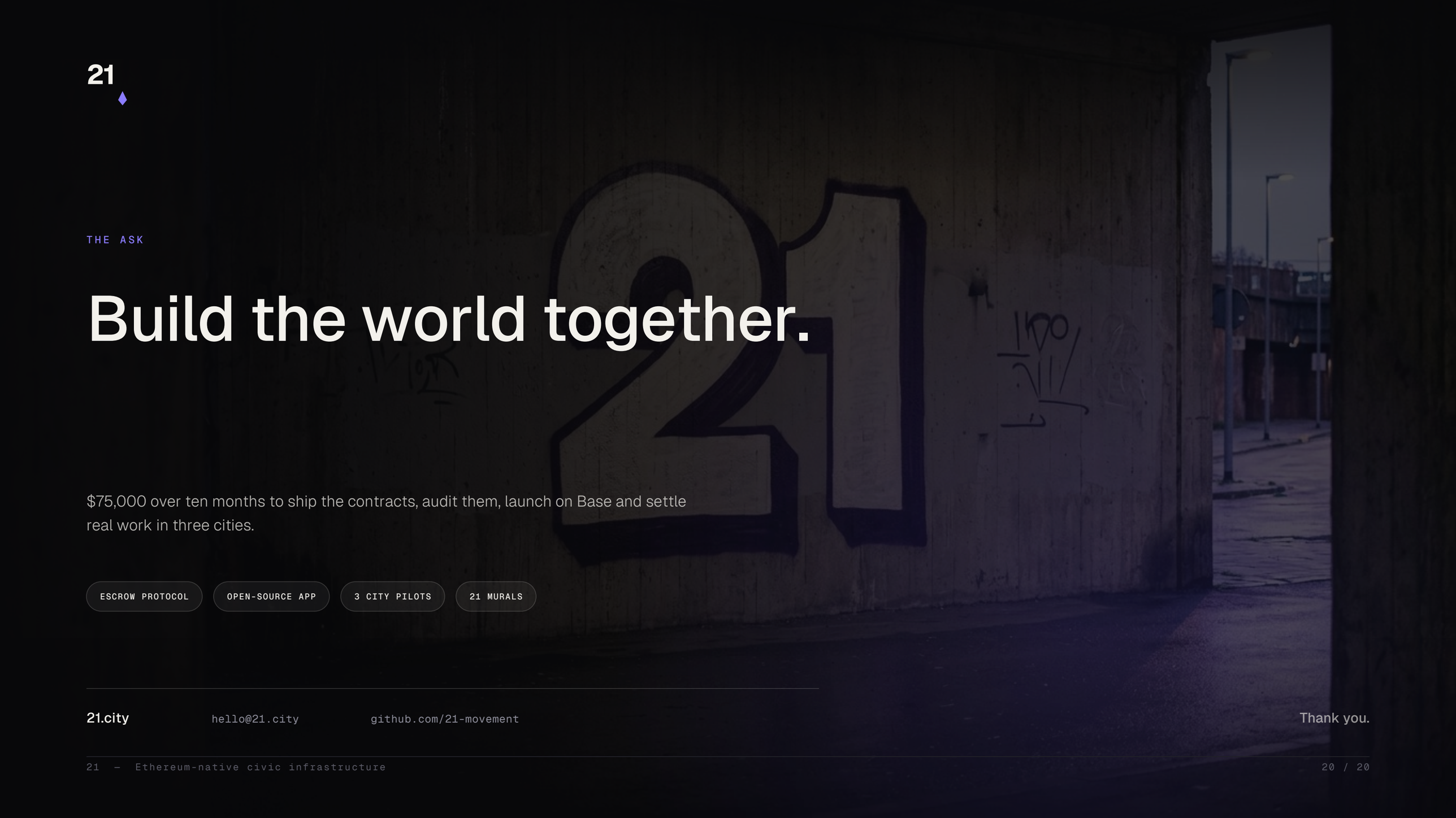
10 cities
EU + one emerging-market municipality

300 tasks
recurring maintenance, not one-off demos

€1m escrow
mostly partner-funded matching pools

1 SDK
any civic org can run their own instance

The registry stays public, the code stays MIT, and the three pilots decide whether this is real.



21

THE ASK

Build the world together.

$75,000 over ten months to ship the contracts, audit them, launch on Base and settle real work in three cities.

ESCROW PROTOCOL · OPEN-SOURCE APP · 3 CITY PILOTS · 21 MURALS

21.city hello@21.city github.com/21-movement

Thank you.